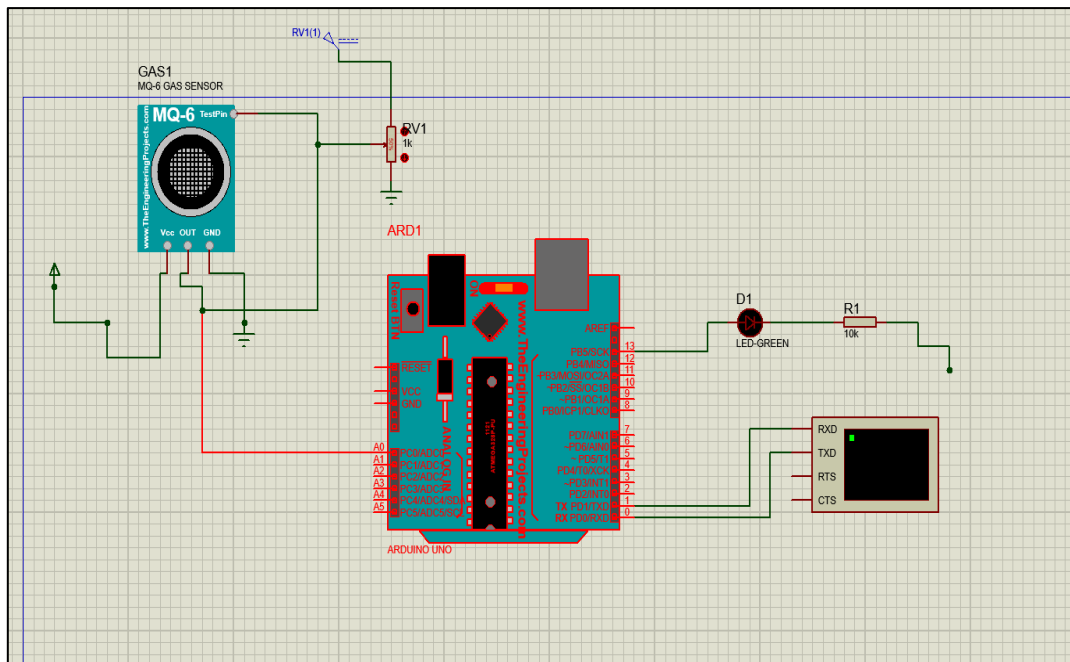


ECA1411-EMBEDDED SYSTEM

LAB MANUAL

Exp No: 8 GAS SENSOR MQ-2 IN PROTEUS USING ARDUINO



```
EXP8 | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
EXP8.ino
1 // Define pin connections
2 const int mq6Pin = A0; // MQ-6 Sensor connected to Analog Pin A0
3 const int alertPin = 13; // Buzzer/LED connected to Digital Pin 13
4
5 // Set the threshold value for gas detection (adjust as needed for simulation)
6 const int threshold = 400;
7
8 void setup() {
9 // Initialize serial communication at 9600 bits per second
10 Serial.begin(9600);
11
12 // Set pin modes
13 pinMode(mq6Pin, INPUT);
14 pinMode(alertPin, OUTPUT);
15 }
16
17 void loop() {
18 // Read the analog value from the MQ-6 sensor (returns 0 to 1023)
19 int sensorValue = analogRead(mq6Pin);
20
21 // Print the data to the Serial Monitor / Virtual Terminal
22 Serial.print("MQ6 Sensor Value: ");
23 Serial.println(sensorValue);
24
25 // Check if gas concentration exceeds the safe threshold
26 if (sensorValue > threshold) {
27 digitalWrite(alertPin, HIGH); // Turn ON buzzer/LED
28 Serial.println("--- ALERT: Gas Detected ---");
29 } else {
30 digitalWrite(alertPin, LOW); // Turn OFF buzzer/LED
31 }
32 }
33
34 Output
35 Sketch uses 2334 bytes (7%) of program storage space. Maximum is 32256 bytes.
36 Global variables use 236 bytes (11%) of dynamic memory, leaving 1812 bytes for local variables. Maximum is 2048 bytes.
```